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ORIGINAL ARTICLE

Plasma ofCS-modified CD44 predicts the survival of patients with lung cancer

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Abstract

Plasma levels of oncofetal chondroitin sulfate (ofCS)-modified CD44 have emerged as a promising biomarker for multi-cancer detection. Here, we explored its potential to predict the survival of patients with lung cancer. A prospective observational cohort was conducted involving 274 newly diagnosed patients with lung cancer at the Sun Yat-sen University Cancer Center from 2013 to 2015. The plasma levels of ofCS-modified CD44 were measured, and Cox regression analysis was performed to assess the association between plasma-modified CD44 levels and overall survival (OS) as well as other prognostic outcomes. Prognostic nomograms were constructed based on plasma ofCS-modified CD44 levels to predict survival outcomes for patients with lung cancer. Patients with high expression ofCS-modified CD44 exhibited significantly worse outcomes in terms of OS (HR = 1.61, 95%CI = 1.13–2.29, $p = 0.009$) and progression-free survival (PFS). These findings were consistent across various analyses. The concordance index of the prognostic nomogram for predicting OS in both the training set and validation set were 0.723 and 0.737, respectively. Additionally, time-dependent receiver operating characteristic (ROC) curves showed that the nomogram could serve as a useful tool for predicting OS in patients with lung cancer. Plasma ofCS-modified CD44 may serve as an independent prognosis marker for patients with lung cancer. Further validation of its predictive value could enhance prognostic assessment and guide personalized treatment strategies for patients with lung cancer.

KEYWORDS
CD44, lung cancer, nomogram, oncofetal chondroitin sulfate, prognosis

Abbreviations: C-index, concordance index; CS, chondroitin sulfate; EMT, epithelial–mesenchymal transition; GAG, glycosaminoglycan; HIS, hospital information system; ofCS, oncofetal chondroitin sulfate; ofCSPGs, ofCS-modified proteoglycans; OS, overall survival; PFS, progression-free survival.

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1 | INTRODUCTION

Lung cancer, a multifaceted malignancy driven by aberrations in numerous biomolecular signaling pathways, ranks among the deadliest forms of cancer, with nearly 1.8 million new cases reported annually.[1] Over recent decades, considerable research efforts have been dedicated to identifying biomarkers for the detection, monitoring, treatment, and prognosis of lung cancer.[2,3]

Chondroitin sulfate is one of the common glycosylation types with a GAG attached to the backbones of specific proteoglycans found on the cell surface of or secreted into the microenvironment surrounding the cells or bodily fluids.[4] CS molecules can vary significantly in length.[5] Oncofetal CS, characterized by a distinct epitope typically limited to trophoblastic cells in the placenta, has been detected in numerous tumors, thereby offering a broad target for human cancer cell identification.[6]

rVAR2, a recombinant fragment derived from *Plasmodium falciparum* VAR2CSA protein, has undergone screening for its affinity to a specific CS oligosaccharide motif,[7–9] rendering it highly specific and with an unprecedented affinity for ofCS.[6] Its binding to cancer cells is notably independent of EMT processes and the cell's epithelial or mesenchymal origin, exhibiting high specificity with minimal to negligible binding in normal tissue, except for the placenta.[6,10,11] Recent studies have harnessed rVAR2 to detect circulating tumor cells from the blood of patients with cancer[10] or to target ofCSPGs in the urine of patients with bladder cancer.[12]

CD44, a stem-like cell receptor, is implicated in various aspects of tumor biology including progression,[13] metastasis,[14] drug resistance[13] and disease prognosis.[15] CD44 expression in cancer stem cells has been correlated with metastasis and resistance to apoptosis. Additionally, it has been reported that ofCS can attach covalently to the backbone of the CD44 protein[6,16] and be secreted into the bloodstream.

Numerous studies have highlighted aberrant glycans as potential biomarkers for cancer prognosis.[17,18] In lung cancer cells, altered glycosylation patterns of glycosphingolipids have been described.[19] Our recent study demonstrated that plasma ofCS-modified CD44 is a promising biomarker for cancer detection, targeted by rVAR2. However, the roles of ofCS-modified CD44 in lung cancer remain largely unclear, with limited reports on its expression and clinical significance in this context. Therefore, this study aimed to explore the associations between ofCS-modified CD44 expression in patients with lung cancer and their clinicopathological characteristics and prognosis. Furthermore, we established a nomogram based on the plasma ofCS-modified CD44 level. These findings are anticipated to pave the way for further investigation into novel diagnostic and prognostic markers of lung cancer, as well as potential therapeutic targets.

2 | MATERIALS AND METHODS

2.1 | Study population and data collection

Patients with lung cancer were selected from the biobank of the Sun Yat-sen University Cancer Center (SYSUCC; Guangzhou, China) based on the following criteria: (1) histological confirmation of malignant tumor, (2) absence of prior clinical treatment such as chemotherapy or radiotherapy before blood collection or surgery, and (3) age under 80 years old. In total, 274 patients with lung cancer diagnosed between January 2013 and December 2015 were included in this study. Clinical and demographic characteristics were retrieved from the HIS of SYSUCC. All patients received surgery, chemotherapy, target therapy or radiotherapy according to the NCCN criteria. This study was approved by the Human Ethics Committee of the SYSUCC (approval number: GZJZ-SB2016-022). Written informed consent was obtained from all participants for the use of their plasma samples.

2.2 | Blood sample collection

The plasma samples were sourced from the SYSUCC biobank. All blood samples were collected following the standard operating procedure. Venous blood (4–5 mL) was drawn into EDTA-containing tubes and immediately centrifuged at 4000 ***g*** for 10 min for plasma collection. Plasma aliquots were stored at −80°C either immediately or within 4 h after blood collection.

2.3 | Enzyme-linked immunosorbent assay

Anti-CD44 was purchased from Proteintech (USA, 60224-1-Ig) and rVAR2 recombinant fragments were expressed and purified as described previously.[20] The sandwich ELISA was developed using anti-CD44 antibodies as the coating proteins, and HRP-labeled rVAR2 as the detecting proteins to quantify the ofCS-modified CD44 in the plasma. The antibodies were diluted in a coating buffer containing 50 mmol/L carbonate buffer, pH 9.6 (C3041, Sigma), and 96-well microplates were coated with 50 μL per/well of the diluted antibodies at 4°C overnight. The coated 96-well plates were washed three times with wash buffer containing 0.05%Tween-20 in TBS (TBST), followed by 200 μL blocking buffer (1% gelatin in TBST) per well for a 2-h incubation at 37°C. After three washes with the washing buffer, 50 μL serum (1:25 dilution) was added and incubated at room temperature. The HRP-labeled recombinant rVAR2 at a final concentration of 0.1 ng/μL was added into the wells after five washes with the wash buffer. Followed by a 1-h incubation at room temperature and five washes with the wash buffer, the plates were supplemented with 100 μL tetramethylbenzidine solution (TMB; KGP125100, KeyGEN). Finally, the reaction was terminated by adding the stop

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solution (KGP12710, KeyGEN) and optical density measurements were taken at 450 nm on a microplate Spectrophotometer (Epoch™ 2, BioTek).

2.4 | Immunohistochemistry

Formalin-fixed, paraffin-embedded lung cancer tissues were collected from the tumor biobank of SYSUCC. The sections were dewaxed in xylene and dehydrated in a serial concentration of ethanol, followed by incubation with 3% H_2O_2 to inactivate the endogenous peroxidase, the sections were then blocked by 5% BSA in PBS with antigen retrieval. rVAR2 were added and incubated overnight at 4°C, followed by incubation with anti-V5 tag antibody and HRP-labeled goat anti-mouse antibody and visualized using a 3,3-diaminobenzidine (DAB) kit (P0203, Beyotime). The positive staining regions were detected using a K-Viewer, an advanced image analysis software, and the staining score was calculated by the DAB-positive segment.

2.5 | Follow-up and outcomes

Patients were primarily followed up via telephone interviews, retrieval of the HIS medical records, and accessing death registration data from the public security bureau. The follow-up period was extended until December 2020. The primary endpoints of this study included death, disease progression, distant metastasis, and recurrence. Distant metastasis and recurrence were defined as the appearance of a newly detected recurrence or distant metastasis, confirmed by imaging diagnosis. Overall survival was determined from the date of diagnosis to either the date of death or to the date of the last follow-up visit. Progression-free survival was defined as the duration from the date of diagnosis to the first day of local recurrence or distant metastasis or to the date of the last follow-up visit.

2.6 | Statistical analysis

The patients were categorized into low to high groups of plasma ofCS-modified CD44 by the median from the training cohort. Next, 60% of patients were randomly assigned to the training set, while the remaining 40% were assigned to the validation set. We classified plasma ofCS CD44 loads into high and low groups with the thresholds of median of training set. Life table estimation was performed using the method of Kaplan–Meier. Univariate comparison of survival curves was performed using the log-rank test. Univariate Cox regression was performed to screen for significant variables. Multivariate Cox proportional hazards model was used to estimate hazard ratios and 95% confidence intervals. Adjusted variables in the model included gender, clinical stage, lymph node metastasis, and remote metastasis that were significant in univariate Cox regression. All statistical tests were two-sided, and p-value <0.05 was considered statistically significant. Analyses were performed using the R 3.6.0 software (R Foundation for Statistical Computing, Vienna, Austria, http://www.rpro ject.org).

The nomograms were constructed to forecast OS status using the findings from the univariate Cox regression analysis. The performance of the nomogram was evaluated by C-index in the training set and validation set. The nomogram calibration curves were used to estimate the agreement between prediction and observation in the probability of 2-, 3- and 5-year survival. Additionally, the discrimination capability of the nomogram was assessed by time-dependent receiver operating characteristic (tdROC) curve analysis.

3 | RESULTS

3.1 | Patient characteristics and follow-up

The baseline characteristics of the patients enrolled in this study are summarized in Table 1. The median age of the population was 58 (interquartile range, IQR = 50–63.75) years, with a male-to-female ratio of 2.1 (186 men and 88 women). In this study, 71 patients (25.9%) were classified as stage I, 36 patients (13.1%) as stage II, 76 patients (27.7%) as stage III and 91 patients (33.2%) as stage IV. The median follow-up time was 41 months (interquartile range, 16–78 months), during which time 125 (45.6%) patients died and 155 (56.6%) experienced disease progression.

Immunohistochemical staining was performed to investigate the expression of ofCS in lung cancer tissue ($N = 9$), the expression levels in tumor tissue were positively correlated with the expression of ofCS-modified CD44 in plasma (Figure S1). The patients were divided into high- or low-ofCS modified CD44 expression groups using a cutoff value of 5.08 relative unit per microliter plasma. The distribution of clinical characteristics did not exhibit significant differences between the two subgroups, except for the smoking status, which might be a risk factor associated with the high expression of ofCS-modified CD44 (Table S1).

3.2 | Survival status and factors associated with OS and PFS of patients with lung cancer

The overall survival (OS) rates of patients with lung cancer at 2 years, 3 years, and 5 years were 73.6%, 61.7%, and 54.9%, respectively. The PFS rates at 2-, 3-, and 5-years were 50.1%, 43.8% and 39.2%, respectively.

Kaplan–Meier survival analysis revealed that, in both the training and validation cohorts, as well as the combined dataset, patients with high ofCS-modified CD44 expression had a significantly shorter OS compared with those with low ofCS-modified CD44 expression. The difference between high and low expression groups was statistically significant ($p = 0.01$, 0.049 and 0.003, respectively; Figure 1A). In addition, high levels of plasma ofCS-modified CD44 were predictive of worse PFS in the training cohort, validation

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TABLE 1 Characteristics of patients with lung cancer in the training, validation and combined datasets.

Characteristic	Training cohort (*N* = 164)	Validation cohort (*N* = 110)	Combined dataset (*N* = 274)
Age			
≤60 years	100 (61.0)	63 (57.3)	163 (59.5)
>60 years	64 (39.0)	47 (42.7)	111 (40.5)
Gender			
Female	52 (31.7)	36 (32.7)	88 (32.1)
Male	112 (68.3)	74 (67.3)	186 (67.9)
Smoking			
Never	85 (51.8)	58 (52.7)	143 (52.2)
Ever	79 (48.2)	52 (47.3)	131 (47.8)
Alcohol			
Never	118 (72.0)	79 (71.8)	197 (71.9)
Ever	46 (28.0)	31 (28.2)	77 (28.1)
Family history			
No	123 (75.0)	89 (80.9)	212 (77.4)
Yes	41 (25.0)	21 (19.1)	62 (22.6)
ALK			
Wild type	83 (50.6)	65 (59.1)	148 (54.0)
Mutation type	3 (1.8)	3 (2.7)	6 (2.2)
Unknown	78 (47.6)	42 (38.2)	120 (43.8)
EGFR			
Wild type	74 (45.1)	55 (50.0)	129 (47.1)
Mutation type	57 (34.8)	42 (38.2)	99 (36.1)
Unknown	33 (20.1)	13 (11.8)	46 (16.8)
KRAS			
Wild type	33 (20.1)	32 (29.1)	65 (23.7)
Mutation type	7 (4.3)	3 (2.7)	10 (3.7)
Unknown	124 (75.6)	75 (68.2)	199 (72.6)
Histology			
Lung squamous cell carcinoma	111 (67.7)	78 (70.9)	189 (69.0)
Lung adenocarcinoma	21 (12.8)	17 (15.5)	38 (13.9)
Else	32 (19.5)	15 (13.6)	47 (17.1)
Differentiation			
Well	15 (9.2)	12 (10.9)	27 (9.9)
Medium	53 (32.3)	40 (36.4)	93 (33.9)
Poor	92 (56.1)	55 (50.0)	147 (53.6)
Unknown	4 (2.4)	3 (2.7)	7 (2.6)
Clinical stage			
STAGE I–II	65 (39.6)	42 (38.2)	107 (39.1)
STAGE III–IV	99 (60.4)	68 (61.8)	167 (60.9)
Lymph node metastasis			
No	54 (32.9)	35 (31.8)	89 (32.5)
Yes	110 (67.1)	75 (68.2)	185 (67.5)
Remote metastasis			
No	111 (67.7)	72 (65.5)	183 (66.8)
Yes	53 (32.3)	38 (34.5)	91 (33.2)

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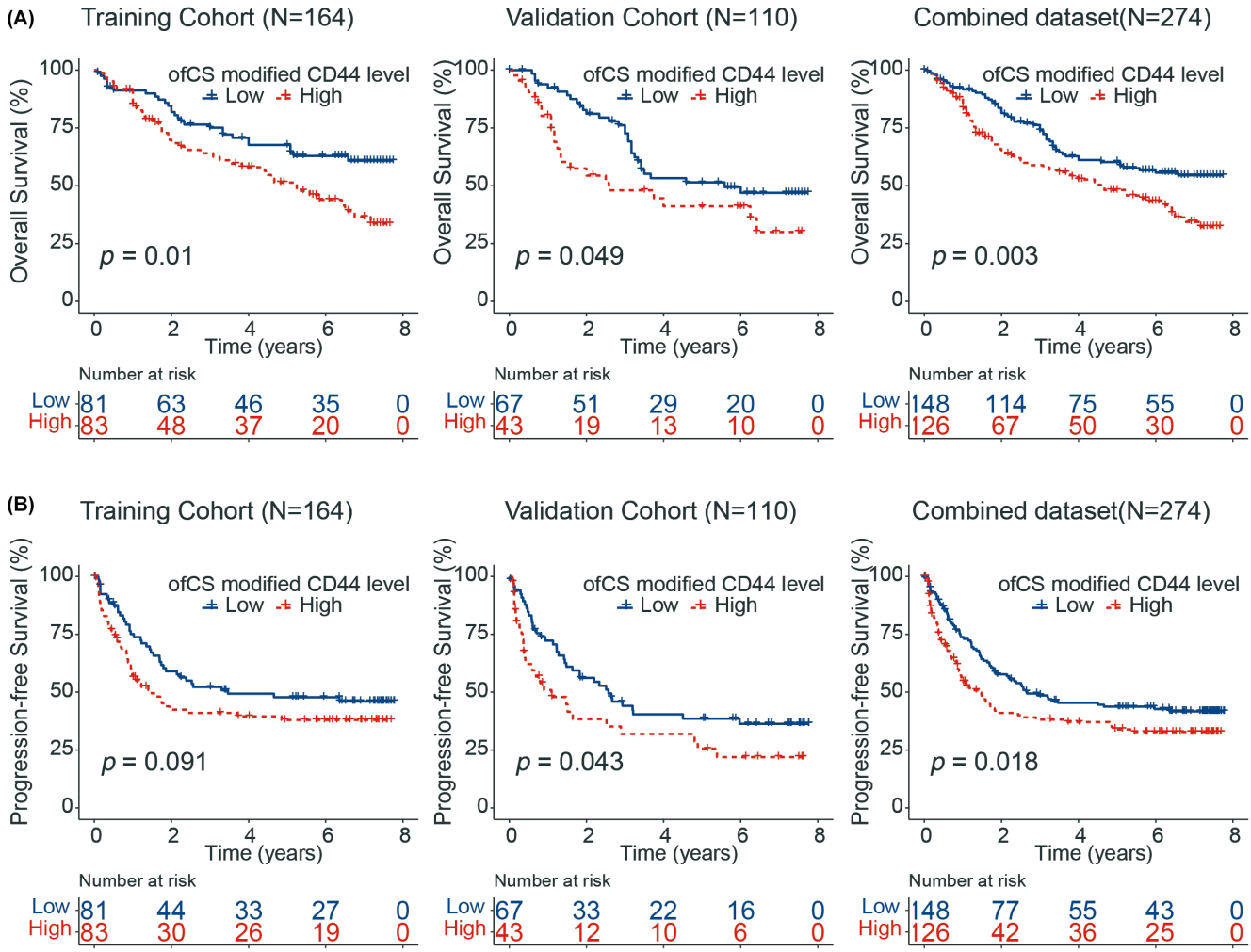


FIGURE 1 Kaplan–Meier survival curves of ofCS-modified CD44 in patients with lung cancer in training cohort, validation cohort and combined dataset for (A) overall survival (OS), (B) progression-free survival (PFS).

cohort, and the combined dataset ($p=0.091$, 0.043, and 0.018, respectively; Figure 1B).

We conducted further analyses on patients with lung cancer who had undergone chemotherapy and surgery separately. The results indicated that, among the 181 patients who received chemotherapy, those with high ofCS-modified CD44 expression had significantly shorter OS ($p=0.026$) and a trend toward shorter PFS ($p=0.098$) compared with those with low ofCS modified CD44 expression (Figure S2A,C). Similarly, among the 182 patients treated with surgery, the OS ($p=0.26$) and PFS ($p=0.07$) of patients with high ofCS-modified CD44 expression were shorter than those with low ofCS-modified CD44 expression (Figure S2B,D).

3.3 | Factors affecting the prognosis of lung cancer

Univariate Cox proportional hazard regression analysis indicated that gender was significantly associated with OS in both the training cohort (HR: 2.03; 95% CI: 1.18–3.52; $p=0.011$) and the validation cohort (HR: 2.20; 95% CI: 1.17–4.13; $p=0.014$). Similarly, the clinical stage was found to be a significant predictor of OS in both cohorts (HR: 2.77; 95% CI: 1.65–4.63; $p<0.001$ for the training cohort, and HR: 3.65; 95% CI: 1.93–6.90; $p<0.001$ for the validation cohort). Lymph node metastasis (HR: 1.95; 95% CI: 1.15–3.31; $p=0.013$, and HR: 3.12; 95% CI: 1.60–6.12; $p=0.001$ for training cohort and validation cohort, respectively) and remote metastasis (HR: 3.58; 95% CI: 2.23–5.75; $p<0.001$, and HR: 4.39; 95% CI: 2.48–7.75; $p<0.001$ for the training cohort and validation cohort, respectively) were also identified as significant risk factors for poor OS. Furthermore, ofCS-modified CD44 expression was associated with poorer OS, although this association was marginally significant in the validation cohort (HR: 1.85; 95% CI: 1.15–2.99; $p=0.012$, and HR: 1.71; 95% CI: 0.99–2.94; $p=0.052$ for training cohort and validation cohort, respectively). These findings are summarized in Table 2.

Moreover, multivariate Cox proportional hazard regression analysis demonstrated that gender (HR = 1.82; 95% CI = 1.19–2.7,

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TABLE 2 Univariate and multivariate Cox regression analysis of prognostic factors in patients with lung cancer for OS.

	Univariate analysis					
	Training cohort		Testing cohort		Combined dataset	
Characteristic	HR (95%CI)	*p*	HR (95%CI)	*p*	HR (95%CI)	*p*
Age						
≤60 years	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
>60 years	0.88 (0.54–1.43)	0.611	0.86 (0.50–1.48)	0.581	0.87 (0.61–1.25)	0.451
Gender						
Female	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
Male	2.03 (1.18–3.52)	0.011	2.20 (1.17–4.13)	0.014	2.09 (1.38–3.16)	<0.001
Smoking						
Never	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
Ever	1.23 (0.77–1.97)	0.376	1.87 (1.09–3.20)	0.023	1.44 (1.01–2.04)	0.043
Alcohol						
Never	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
Ever	1.08 (0.65–1.80)	0.758	0.80 (0.43–1.47)	0.473	0.96 (0.65–1.42)	0.842
Family history						
No	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
Yes	0.71 (0.40–1.26)	0.248	1.57 (0.86–2.90)	0.145	0.97 (0.64–1.47)	0.885
Histology						
Lung squamous cell carcinoma	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
Lung adenocarcinoma	3.29 (1.79–6.07)	<0.001	2.39 (1.18–4.83)	0.016	2.83 (1.78–4.48)	<0.001
Else	1.23 (0.68–2.26)	0.494	0.84 (0.35–1.99)	0.69	1.05 (0.64–1.72)	0.839
Differentiation						
Well	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
Medium	2.47 (0.75–8.17)	0.138	0.86 (0.34–2.15)	0.74	1.40 (0.69–2.87)	0.353
Poor	2.70 (0.84–8.73)	0.097	1.05 (0.44–2.53)	0.912	1.58 (0.79–3.17)	0.194
Unknown	2.54 (0.26–24.5)	0.421	0.00 (0.00-.)	0.997	0.62 (0.08–4.91)	0.652
ALK						
Wild type	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
Mutation type	0.65 (0.09–4.74)	0.67	0.52 (0.07–3.83)	0.525	0.60 (0.15–2.43)	0.47
Unknown	1.27 (0.80–2.04)	0.313	0.92 (0.52–1.63)	0.772	1.09 (0.77–1.56)	0.625
EGFR						
Wild type	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
Mutation type	1.09 (0.64–1.86)	0.739	1.10 (0.62–1.94)	0.745	1.10 (0.75–1.62)	0.635
Unknown	1.75 (0.95–3.23)	0.072	1.05 (0.43–2.57)	0.909	1.45 (0.88–2.38)	0.145
KRAS						
Wild type	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
Mutation type	0.31 (0.04–2.38)	0.261	0.48 (0.06–3.55)	0.47	0.33 (0.08–1.35)	0.122
Unknown	1.19 (0.66–2.13)	0.566	0.69 (0.40–1.19)	0.181	0.88 (0.59–1.30)	0.51
Clinical stage						
STAGE I–II	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
STAGE III–IV	2.77 (1.65–4.63)	<0.001	3.65 (1.93–6.90)	<0.001	3.09 (2.07–4.60)	<0.001
Lymph node metastasis						
No	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
Yes	1.95 (1.15–3.31)	0.013	3.12 (1.60–6.12)	0.001	2.37 (1.56–3.57)	<0.001

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TABLE 2 (Continued)

Characteristic	Univariate analysis					
	Training cohort		Testing cohort		Combined dataset	
	HR (95%CI)	p	HR (95%CI)	p	HR (95%CI)	p
Remote metastasis						
No	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
Yes	3.58 (2.23–5.75)	<0.001	4.39 (2.48–7.75)	<0.001	3.77 (2.63–5.42)	<0.001
ofCS-CD44						
Low	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
High	1.85 (1.15–2.99)	0.012	1.71 (0.99–2.94)	0.052	1.69 (1.19–2.41)	0.003

$p=0.006$), remote metastasis (HR = 2.91; 95% CI = 1.88–4.52, $p<0.001$), and ofCS-modified CD44 expression (HR = 1.61; 95% CI = 1.13–2.29, $p=0.009$) were independent risk factors for poor prognosis in the combined dataset (Table 2).

In addition, univariate Cox proportional hazard regression analysis indicated that gender (HR: 1.49; 95% CI: 0.94–2.37; $p=0.091$, and HR: 2.03; 95% CI: 1.18–3.50; $p=0.011$ for training cohort and validation cohort, respectively), clinical stage (HR: 1.91; 95% CI: 1.23–2.97; $p=0.004$, and HR: 2.45; 95% CI: 1.43–4.19; $p=0.001$ for training cohort and validation cohort, respectively), lymph node metastasis (HR: 1.62; 95% CI: 1.02–2.57; $p=0.043$, and HR: 3.27; 95% CI: 1.77–6.03; $p<0.001$ for training cohort and validation cohort, respectively), and remote metastasis (HR: 1.84; 95% CI: 1.19–2.86; $p=0.006$, and HR: 1.98; 95% CI: 1.20–3.26; $p=0.007$ for training cohort and validation cohort, respectively), as well as plasma ofCS-modified CD44 (HR: 1.43; 95% CI: 0.94–2.18; $p=0.093$, and HR: 1.65; 95% CI: 1.01–2.68; $p=0.045$ for training cohort and validation cohort, respectively) were associated with poor PFS (Table 3). However, multivariate analyses showed that only gender remained an independent risk factor for the PFS, suggesting that although ofCS-modified CD44 was associated with PFS, this relationship did not reach statistical significance.

3.4 | Nomogram development with plasma ofCS-modified CD44

To assess the clinical prognosis value of plasma ofCS-modified CD44, prognostic nomograms were constructed using the training set to predict the 2-, 3- and 5-year OS based on variables of plasma ofCS-modified CD44, gender, clinical stage, lymph node metastasis, remote metastasis, (Figure 2A). The calibration plot for the probability of 2-, 3- and 5-year OS revealed favorable agreement between nomogram prediction and actual observation in both the training cohort, validation cohort, and combined dataset (Figure 2B). The C-index for OS in the training set and validation set were 0.723 and 0.737, respectively, indicating the nomogram with plasma ofCS-modified CD44 had a good prognostic stratification.

To assess the predictive value of the model, a tdROC curve was created. As shown in Figure S3, the areas under the curve (AUCs) of the risk score in the training cohort at 2 years, 3 years, and 5 years were 0.839, 0.860, and 0.775, respectively. In the validation cohort, the AUCs were 0.710, 0.831, and 0.730 and, in the combined dataset, they were 0.806, 0.852, and 0.765, respectively. These findings further underscored the predictive utility of the nomogram incorporating plasma ofCS-modified CD44 for assessing patient outcomes over time.

4 | DISCUSSION

Lung cancer stands as the leading cause of cancer death worldwide.[1] Despite the advances in early diagnosis and treatment for lung cancer, the OS rate of patients with lung cancer remains less than 20%.[1] Therefore, there is an urgent need to explore effective markers to guide clinical diagnosis and individualized treatment. Given the intricacies and heterogeneity of lung cancer,[21] the quest for a singular molecular profile for clinical prognostication is fraught with complexity and challenges. Although precision diagnosis or progression prediction using gene expression profiling has emerged,[22–24] their high cost and limited reproducibility restrict its widespread application. In this study, we unveil a novel role of plasma ofCS-modified CD44 as a marker for tumor progression predictor of lung cancer.

A recent study showed that aberrant overexpression of CD44 in the metastatic tumor cells relative to normal counterparts, implicating CD44 in metastasis by modulating tumor cell trafficking.[25] However, the involvement of ofCS-modified CD44 in the occurrence and development of lung cancer remains largely unknown. In this study, we pioneered investigations into ofCS-modified CD44, a specific protein modification, and interrogated its expression in serum from human patients with lung cancer, probing its association with clinicopathological parameters. Our findings confirmed that elevated plasma levels of ofCS-modified CD44 were not linked to clinical stages and lymph node metastasis in patients with lung cancer, but rather correlated with the smoking status and the occurrence of remote metastasis. As reported previously, chondroitin sulfate

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TABLE 3 Univariate and multivariate Cox regression analysis of prognostic factors in patients with lung cancer for PFS.

	Univariate analysis						Multivariate analysis	
	Training cohort		Testing cohort		Combined dataset		Combined dataset	
Characteristic	HR (95%CI)	*p*	HR (95%CI)	*p*	HR (95%CI)	*p*	HR (95%CI)	*p*
Age								
≤60 years	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.		
>60 years	0.69 (0.45–1.08)	0.105	0.89 (0.54–1.45)	0.633	0.78 (0.56–1.08)	0.134		
Gender								
Female	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
Male	1.49 (0.94–2.37)	0.091	2.03 (1.18–3.50)	0.011	1.68 (1.18–2.39)	0.004	1.46 (1.01, 2.10)	0.043
Smoking								
Never	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.		
Ever	0.91 (0.60–1.39)	0.677	1.43 (0.88–2.32)	0.146	1.08 (0.79–1.49)	0.615		
Alcohol								
Never	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.		
Ever	1.08 (0.68–1.71)	0.751	1.29 (0.77–2.18)	0.338	1.15 (0.81–1.63)	0.425		
Family history								
No	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.		
Yes	0.98 (0.61–1.58)	0.936	1.46 (0.83–2.57)	0.185	1.11 (0.77–1.60)	0.572		
Histology								
Lung squamous cell carcinoma	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.		
Lung adenocarcinoma	1.03 (0.53–2.01)	0.936	1.84 (0.97–3.50)	0.061	1.33 (0.84–2.11)	0.219		
Else	1.02 (0.59–1.77)	0.945	1.32 (0.62–2.80)	0.476	1.08 (0.70–1.69)	0.718		
Differentiation								
Well	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.		
Medium	1.47 (0.56–3.83)	0.43	1.64 (0.63–4.30)	0.311	1.54 (0.78–3.03)	0.212		
Poor	2.24 (0.90–5.61)	0.084	2.12 (0.83–5.42)	0.117	2.14 (1.11–4.12)	0.023		
Unknown	4.96 (1.18–20.8)	0.029	2.72 (0.53–14.1)	0.232	3.76 (1.28–11.0)	0.016		
ALK								
Wild type	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.		
Mutation type	1.02 (0.25–4.21)	0.976	0.42 (0.06–3.03)	0.386	0.69 (0.22–2.19)	0.53		
Unknown	1.01 (0.66–1.54)	0.969	1.27 (0.77–2.08)	0.344	1.08 (0.78–1.49)	0.644		
EGFR								
Wild type	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.		
Mutation type	1.01 (0.63–1.60)	0.982	0.95 (0.57–1.58)	0.845	0.99 (0.70–1.39)	0.951		
Unknown	0.90 (0.50–1.63)	0.728	1.56 (0.72–3.40)	0.262	1.02 (0.64–1.63)	0.939		
KRAS								
Wild type	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.		
Mutation type	0.60 (0.18–2.03)	0.413	0.86 (0.11–6.46)	0.88	0.68 (0.24–1.92)	0.469		
Unknown	0.95 (0.58–1.58)	0.855	1.67 (0.95–2.93)	0.076	1.20 (0.82–1.74)	0.346		
Clinical stage								
STAGE I–II	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
STAGE III–IV	1.91 (1.23–2.97)	0.004	2.45 (1.43–4.19)	0.001	2.11 (1.51–2.97)	<0.001	1.15 (0.67, 1.98)	0.614

(Continues)

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TABLE 3 (Continued)

	Univariate analysis						Multivariate analysis	
	Training cohort		Testing cohort		Combined dataset		Combined dataset	
Characteristic	HR (95%CI)	*p*	HR (95%CI)	*p*	HR (95%CI)	*p*	HR (95%CI)	*p*
Lymph node metastasis								
No	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
Yes	1.62 (1.02–2.57)	0.043	3.27 (1.77–6.03)	<0.001	2.11 (1.46–3.04)	<0.001	1.59 (0.95, 2.68)	0.077
Remote metastasis								
No	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
Yes	1.84 (1.19–2.86)	0.006	1.98 (1.20–3.26)	0.007	1.90 (1.37–2.64)	<0.001	1.44 (0.98, 2.12)	0.067
ofCS–CD44								
Low	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
High	1.43 (0.94–2.18)	0.093	1.65 (1.01–2.68)	0.045	1.46 (1.06–2.00)	0.019	1.36 (0.99, 1.87)	0.06

can be excreted in urine and cover up the clinical stage-dependent manner,[26] in addition, cigarette smoke extract can increase the expression of CS matrix proteoglycan,[27] and its mechanism needs to be further explored.

Alterations in glycosyltransferase levels and glycosylation patterns have been evidenced in inflammatory conditions, tumorigenesis, and metastasis.[28] Changes in glycosyltransferase gene expression have been correlated with the outcome of patients with lung cancer. For instance, high expression of CHST11, an enzyme specifically required for CSA 4-O-sulfation, was reported to be significantly correlated with poor relapse-free survival in three independent lung cancer cohorts.[6] Glycosylation of proteins generates changes in their biophysical properties, function, distribution, and retention in the plasma membrane and modulates cell behavior, cellular interactions, specific ligand–receptor interactions, and immune recognition.[29–31] Collectively, these findings substantiate the notion that aberrant glycan production could serve as a direct prognostic marker.

Nomograms, practical models used in routine laboratory settings, facilitate the prediction of a specific outcome, such as recurrence, based on the characteristics of patients. Currently, numerous nomograms have been proven to be effective in lung cancer, such as the noncytotoxic chemosensitizer model and the epidermal growth factor receptor (EGFR) mutation model.[32,33] In our pursuit of reliable and clinically valuable prognostic models, we devised a nomogram integrating variables gleaned from univariant Cox analysis—gender, clinical stage, lymph node metastasis, remote metastasis, and the expression level of the promising prognosis predictor biomarker—ofCS-modified CD44. This nomogram enables personalized survival risk assessment and shows satisfactory discrimination. In addition, we used a validation cohort to verify the discrimination ability and stability of this model. Such validation is pivotal, ensuring nomogram universality and guarding against overestimation of predictability.[34] The results showed that this nomogram had consistency between prognosis prediction and actual observation, corroborated by robust performance on time-dependent ROC curves. Therefore, the nomogram established in this study provided a robust predictive tool for assessing OS in lung cancer.

Acknowledging certain limitations, our study adopts a retrospective, single-center design, suggesting the need for future multicenter investigations. Furthermore, patients experienced different anticancer treatments and exhibited varying tumor progression, medical treatment responses, and nutritional statuses- factors not comprehensively addressed. Moreover, this observation study leaves unanswered queries regarding how specific CD44 modifications or their combination impact gene expression or tumor behavior, underscore the need for further mechanistic exploration.

In summary, our study highlights the potential significance of plasma ofCS-modified CD44 as an important independent prognostic marker for patients with lung cancer. The nomogram developed in this study, incorporating plasma ofCS-modified CD44, exhibits strong discriminatory power in predicting OS. These findings underscore the utility of plasma ofCS-modified CD44 as a promising biomarker for prognostication in lung cancer, offering valuable insights for clinical decision-making and personalized patient care.

AUTHOR CONTRIBUTIONS

Zi-Yi Wu: Conceptualization; methodology; supervision; writing – original draft. **Da-Wei Yang:** Data curation. **Yong-Qiao He:** Data curation. **Tong-Min Wang:** Data curation. **Ting Zhou:** Data curation. **Xi-Zhao Li:** Data curation. **Pei-Fen Zhang:** Data curation. **Wen-Qiong Xue:** Data curation. **Jiang-Bo Zhang:** Data curation. **Jianbing Mu:** Conceptualization; methodology; supervision; writing – review

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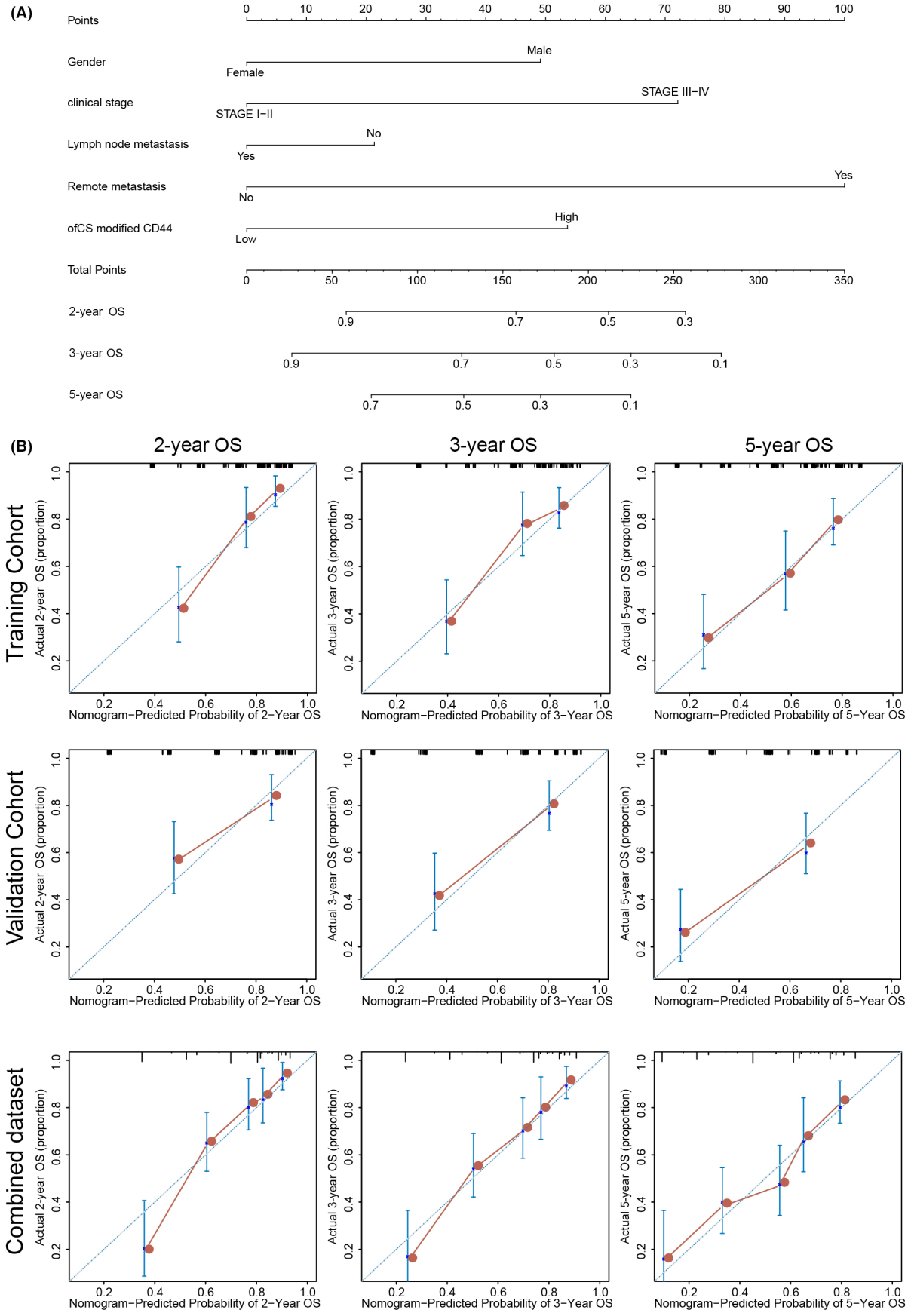


FIGURE 2 Nomogram (A), including age, gender, lymph node metastasis, remote metastasis, plasma ofCS-modified CD44, for 2-, 3- and 5-year overall survival (OS) in the training cohort. Calibration curves (B) of the nomogram for predicting overall survival (OS).

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and editing. **Wei-Hua Jia:** Conceptualization; funding acquisition; investigation; writing – review and editing.

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CONFLICT OF INTEREST STATEMENT

No potential conflicts of interest are disclosed.

DATA AVAILABILITY STATEMENT

All data included in this study are available upon request by contact with the corresponding author.

ETHICS STATEMENT

Approval of the research protocol by an Institutional Reviewer Board: This study was approved by the Human Ethics Committee of the SYSUCC (approval number: GZJZ-SB2016-022).

Informed Consent: Written informed consent was obtained from all participants for the use their plasma samples.

Registry and the Registration No. of the study/trial: N/A.

Animal Studies: N/A.

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